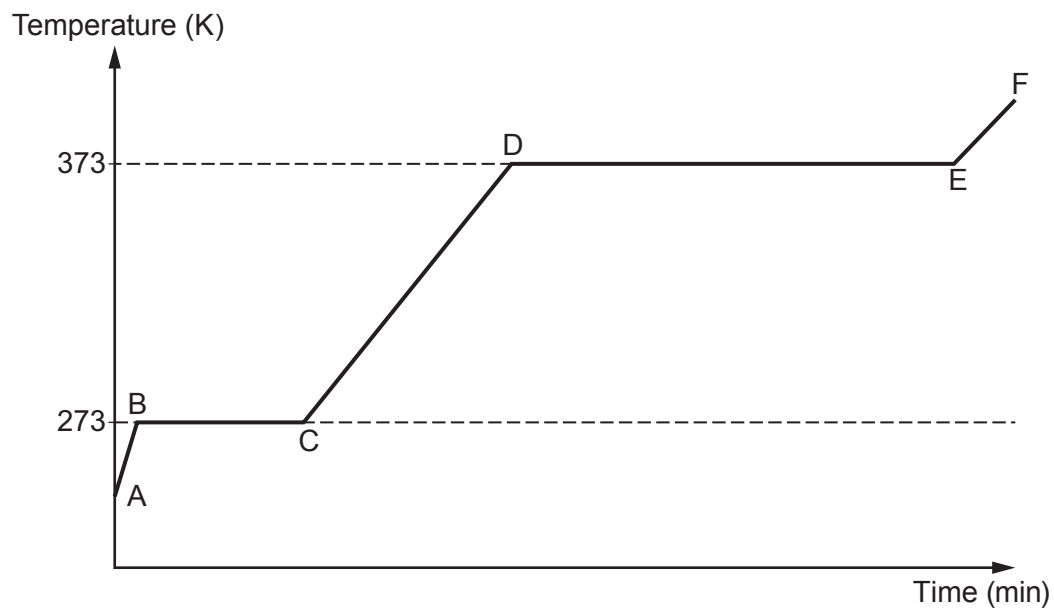


Examiner
only

6. The graph below shows how the temperature of a block of ice changes over time as it is heated.



- (a) Refer to the graph to explain the changes in temperature and changes in state of the block of ice in terms of the behaviour of molecules. [6 QER]



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(b) The specific latent heat, L , of fusion of ice is 336 000 J/kg.

(i) Explain what is meant by this statement. [2]

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(ii) The ice block has a mass of 800 g. Its initial temperature is 250 K.
Use equations from page 2 to calculate the energy required to raise its temperature to 273 K and completely melt the ice block at 273 K.
The specific heat capacity, c , of ice is 2030 J/kg K. [5]

Energy = J

13

